Non-Generative Learning to Rerank for MS/MS-Based Molecule Retrieval

Abstract. Retrieving molecular structures from tandem mass spectra is difficult because many candidates share mass, formula, and substructure information. Joint embedding methods rank by spectrum–molecule similarity, whereas generative retrieval may introduce an autoregressive molecular decoder. We study direct, non-generative learning to rerank a retrieved molecular list. SpecEmbedding-Rerank first aligns a peak-sequence spectrum encoder with a molecular graph encoder. Its second-stage residual ranker combines spectrum and candidate embeddings, explicit interaction features, the base score, and the base rank under listwise supervision. Using the overlap-clean alignment-seed-42 checkpoint and caches, we compare a pointwise scorer without learned cross-candidate interaction with a candidate-set Transformer across three reranker initializations; a separate analysis repeats the design for alignment seeds 42–44. On the 17,556-query MassSpecGym test split, the fixed-alignment mean ± sample-SD Recall@1 values are 67.44 ± 0.74 and 68.58 ± 0.20 for mass-conditioned retrieval, and 73.97 ± 0.59 and 74.57 ± 0.28 for formula-conditioned retrieval, compared with fixed-base values of 47.46 and 63.10. Recall and MRR use supplied candidate files with a local exact-target-SMILES single-positive protocol; the unquantified difference from reference two-dimensional InChIKey identity, which may yield multiple positives, means they are not equivalent to official evaluator outputs. Across the three alignment seeds, both learned rankers improve their corresponding base in all 12 alignment–candidate–learned-model aggregate cells for both Recall@1 and MRR, whereas the Transformer–pointwise direction varies by alignment. A fixed-alignment audit finds that the full model does not achieve six-of-six MRR wins against any of five removal variants; the rank-free variant improves both metrics in every pool-by-reranker-seed pair. Thus, the evidence supports supervised non-generative second-stage reranking; robust self-attention and independent component benefits remain unestablished. Neither variant generates molecules or forces the target into the test top-40 list.

Keywords: Data mining for bioinformatics · Tandem mass spectrometry · Molecule retrieval · Learning to rank · Multimodal representation learning

1 Introduction

MS/MS is widely used to identify small molecules in metabolomics, natural-product discovery, and related biomedical applications. An observed spectrum is only a partial, condition-dependent view of a molecular structure: different molecules can yield similar fragments, while the same molecule can produce different spectra across instruments, adducts, and collision energies. Consequently, structure annotation is naturally a large-scale data mining problem in which a query spectrum must retrieve and rank a correct structure from a chemically constrained candidate pool.

MassSpecGym provides structure-disjoint data splits and standardized mass- and formula-conditioned candidate pools for this problem [3]. Recent methods mainly address the cross-modal gap between spectra and molecules. JESTR learns a spectrum–molecule joint space with contrastive multiview coding and ranks candidates directly by cosine similarity [13,21]. GLMR instead retrieves contextual candidates, generates a refined molecule with a conditional language model, and reranks candidates by molecule–molecule similarity [25].

Both approaches leave an important data mining question open. Candidate-wise cosine scoring does not explicitly train a second-stage listwise objective over the retrieved set, while generative retrieval adds an autoregressive decoder and makes ranking depend on the quality and validity of a generated structure. We ask whether a supervised second-stage ranker can directly separate hard candidates and whether explicit candidate interaction adds value beyond a pointwise residual scorer.

Building on the peak-sequence spectrum architecture introduced by SpecEmbedding [23], we construct SpecEmbedding-Rerank, a two-stage retrieval and reranking framework. The first stage trains that spectrum tower from scratch and aligns it with a GINE molecular graph encoder. The second stage operates only on the first-stage top-K list. It constructs explicit spectrum–candidate interaction features, injects the original score and rank as retrieval priors, and predicts a residual correction to the base score. We instantiate this framework with pointwise and candidate-set Transformer variants.

Our contributions are:

- We formulate and evaluate direct, non-generative second-stage learning-to-rank for MS/MS-to-molecule retrieval, avoiding both cosine-only final ranking and autoregressive molecular decoding.
- We compare pointwise and self-attentive rankers with matched features, score heads, objectives, and optimization. On the same fixed caches, we then audit five removal settings of the full configuration.
- Using the MassSpecGym split and candidate files under our local exact-target-SMILES single-positive protocol, both implementations improve their fixed base retriever across alignment seeds 42–44. The candidate-set Transformer advantage is not directionally uniform across these checkpoints, so the analysis does not establish a robust self-attention benefit. Moreover, within the overlap-clean alignment-seed-42 audit, the full model does not achieve six-of-six pool-by-reranker-seed MRR wins against any removal variant.

AI-assistance disclosure. OpenAI Codex assisted throughout all sections with drafting and language revision, and during the project with code editing, experiment orchestration, and consistency audits. Reported observations and numerical results came from the stated software pipeline, not model generation. The authors reviewed all AI-assisted text and code, verified the technical claims, citations, saved artifacts, and reported results, and assume full responsibility for the manuscript.

2 Related Work

2.1 Annotation Paradigms

Computational MS/MS annotation differs in retrieved objects and ranking evidence. Candidate-score correction predates our work: MetFusion combines insilico fragmentation with spectral-library evidence [8]; MS2Query reranks its top 2,000 MS2DeepScore matches with a five-feature random forest [12]; and LC-MS2Struct combines an MS/MS scorer with retention order to jointly rank candidates from one LC analysis [1]. We instead rank structures in query-specific MassSpecGym pools from frozen spectrum–molecule representations, without spectral-library-neighbor or retention-time evidence [3].

Within structure-database retrieval, CSI:FingerID and MIST predict molecular fingerprints from spectra [7,9]. CMSSP and JESTR learn joint spectrum–molecule spaces; MVP extends contrastive alignment across molecular graphs, fingerprints, individual spectra, and consensus spectra [5,13,6]. These methods order candidates by predicted-fingerprint scores or cross-modal similarity. SpecEmbedding instead studies spectrum–spectrum representation learning [23]; we reuse only its peak-sequence architecture, train it from scratch against molecular graphs, and learn a complementary second-stage order within a retrieved hard-candidate list.

Other paradigms use different intermediate objects. MassFormer predicts tandem spectra from molecular graphs [24], while MSNovelist generates structures from MS/MS-derived fingerprints [20]. GLMR conditions its molecular decoder on retrieved candidates and reranks them by similarity to the generated molecule [25]. Our closed-set method does not decode a structure or return a molecule outside the first-stage list; it directly optimizes the scores of available candidates.

2.2 Learning to Rank Candidate Sets

Learning-to-rank objectives define supervision over preferences or complete lists rather than treating similarity sorting as a fixed post-processing step. RankNet learns pairwise preferences, whereas ListNet defines a probability model over a ranked list [2,4]. Ranking functions can also exploit relations among competing items. Set Transformer provides attention blocks for set-structured inputs [15], while SetRank uses self-attention to model cross-document interactions [17]. We apply these principles to cached cross-modal embeddings with explicit spectrum–candidate pair features and first-stage score and rank priors. Table 1 positions the resulting query-specific, non-generative second stage among selected related MassSpecGym approaches. The contribution is neither reranking nor self-attention itself; residual scoring and candidate interaction are evaluated design choices rather than assumed sources of gain.

Table 1. Mechanistic comparison with selected related MassSpecGym approaches.

Model	Ranking signal	Candidate interaction	Generates	Training stages
JESTR	Spectrum–candidate cosine	None at inference	No	InfoNCE plus formula-candidate regularizer
GLMR	Generated–candidate cosine	Retrieved candidates condition the generator	Yes	Dual-path InfoNCE plus generation
Ours	Residual score from pair features and retrieval priors	Optional candidate self-attention	No	Cross-modal InfoNCE plus listwise/pairwise reranking

3 Problem Formulation

Let s be an MS/MS spectrum, m^+ its correct molecule, and $\mathcal{D}_s$ a query-specific mass- or formula-conditioned candidate pool. A base retriever assigns

$$b(s, m) = \mathrm{sim}(f_s(s), f_m(m)) \tag{1}$$

to every $m \in \mathcal{D}_s$ and returns the top-K list $\mathcal{C}_s = (m_1, \ldots, m_K)$. The reranker learns scores $g(s, m_i, \mathcal{C}_s)$ such that m^+ is ordered as high as possible. Because it cannot return a molecule outside $\mathcal{C}_s$, end-to-end Recall@k is upper-bounded by

$$U_K = \frac{1}{N} \sum_{n=1}^{N} \mathbb{I}[m_n^+ \in \mathcal{C}_{s_n}]. \tag{2}$$

Define the within-list top-k accuracy conditional on first-stage coverage as

$$A_{k|K} = \frac{\sum_{n=1}^{N} \mathbb{I}[m_n^+ \in \mathcal{C}_{s_n},\, \mathrm{rank}_g(m_n^+) \leq k]}{\sum_{n=1}^{N} \mathbb{I}[m_n^+ \in \mathcal{C}_{s_n}]}, \qquad \mathrm{Recall@}k = U_K A_{k|K} \leq U_K. \tag{3}$$

A reranker applied to a fixed cache can therefore change only $A_{k|K}$, not U_K. We report metrics over all queries, assigning zero credit when the positive is absent rather than conditioning the primary metric on successful first-stage retrieval.

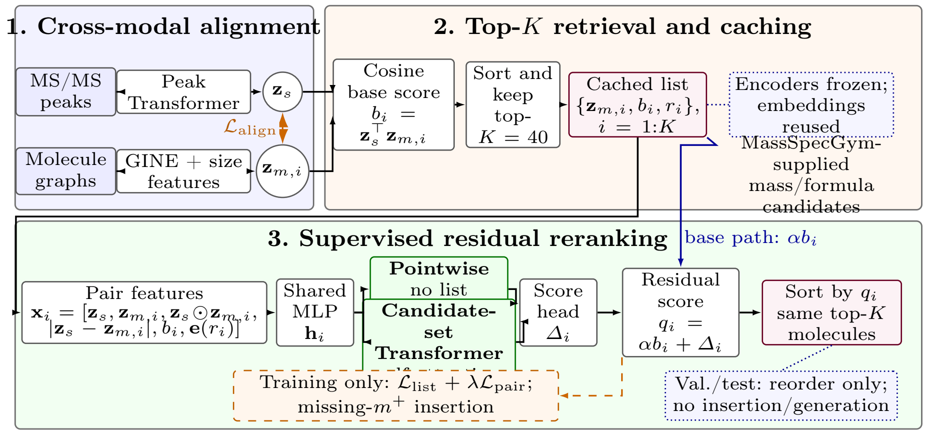

Fig. 1. Overview of SpecEmbedding-Rerank. Panel 1 aligns the spectrum Transformer and molecular GINE. Panel 2 uses their frozen, cached embeddings to produce the top-40 list. Panel 3 applies shared pair features with either pointwise scoring or candidate-set self-attention and adds a residual to the base score. Dashed arrows denote training-only operations. Positives are never forced into the truncated top-40 list at validation/test, and no molecule is generated.

4 Method

4.1 Overview

The framework has three steps: cross-modal alignment, top-K retrieval, and supervised residual reranking. Spectrum and molecule encoders are frozen while the reranker is trained, so their embeddings can be computed once and reused.

4.2 Cross-Modal Base Retriever

The spectrum encoder follows SpecEmbedding's peak-sequence architecture [23], which processes peak m/z values and intensities with sinusoidal m/z features and a Transformer. In our implementation, each spectrum has at most 100 tokens: a precursor-m/z token with intensity 2, followed by up to 99 fragments selected by intensity. Selected fragments retain their source-array order, and their intensities are normalized by the largest selected fragment intensity. Only these m/z–intensity tokens enter the spectrum encoder; adduct, instrument, and collision-energy metadata are not explicit inputs. We initialize this tower from scratch and train it against molecules rather than spectra. The molecular encoder is a GINE network over atom and bond features [11]. Mean-pooled graph features are augmented with explicit graph-size features. Two projection heads map the encoders to a common d-dimensional space:

$$\mathbf{z}_s = \frac{p_s(f_s(s))}{\|p_s(f_s(s))\|_2}, \qquad \mathbf{z}_m = \frac{p_m(f_m(m))}{\|p_m(f_m(m))\|_2}. \tag{4}$$

We use a bidirectional cross-modal contrastive loss that permits multiple positives. Spectra and molecules sharing a molecular label form positives, which avoids treating replicate spectra as false negatives. For a minibatch of size B, let $P(i)$ be the indices sharing the label of item i. With a learned inverse-temperature scale a, the base logits are

$$\ell_{ij} = a\,\mathbf{z}_{s_i}^{\top}\mathbf{z}_{m_j}. \tag{5}$$

The spectrum-to-molecule term is

$$\mathcal{L}_{s\to m} = -\frac{1}{B}\sum_{i=1}^{B}\frac{1}{|P(i)|}\sum_{j\in P(i)}\log\frac{\exp(\ell_{ij})}{\sum_{k=1}^{B}\exp(\ell_{ik})}, \tag{6}$$

and the molecule-to-spectrum term applies the same expression to $\ell^{\top}$. The alignment loss is their mean. The implementation parameterizes $a = \exp(u)$ and caps it at 100. At inference, the normalized dot product $b_i = \mathbf{z}_s^{\top}\mathbf{z}_{m_i}$ is the base score.

4.3 Candidate Construction and Training Protocol

MassSpecGym supplies one candidate pool per query. The base retriever scores the pool and keeps its top $K = 40$ molecules. If a training target is absent from the retrieved top-40, it replaces the last molecule while its full-pool base rank is cached. That rank is one plus the number of strictly higher-scoring candidates; ties with the target do not count as higher. The rank embedding receives $\bar{r}_i = \min\{\max(r_i, 0), 512\}$, with index 0 reserved for padding. Training also shuffles candidate order while moving each candidate's embedding, score, and rank together to guard against cache-order artifacts.

Scored pools are exact-SMILES-deduplicated unions of the labeled target and supplied candidates. A static audit found the target already present in all 32,010 frozen pools, so no pool was enlarged. Post-retrieval positive forcing is training-only; validation and test perform no top-40 replacement. A target absent from the retrieved top-40 remains absent and lowers the upper bound and all end-to-end metrics. No test record, target, or label is used for optimization or checkpoint selection.

4.4 Residual Reranker Variants

For candidate m_i with base rank r_i, we construct

$$\mathbf{x}_i = [\mathbf{z}_s,\ \mathbf{z}_{m_i},\ \mathbf{z}_s \odot \mathbf{z}_{m_i},\ |\mathbf{z}_s - \mathbf{z}_{m_i}|,\ b_i,\ \mathbf{e}(\bar{r}_i)], \tag{7}$$

where $\mathbf{e}(\bar{r}_i)$ is a learned rank embedding. The direct embeddings retain modality-specific evidence, while product and absolute-difference terms make symmetric matching cues explicit. An MLP projects each vector:

$$\mathbf{h}_i = \mathrm{MLP}(\mathbf{x}_i). \tag{8}$$

Both variants receive only the frozen spectrum embedding, frozen candidate embeddings, cosine base scores, clipped base-rank embeddings, and the candidate padding mask. Raw peaks, SMILES tokens, precursor mass or formula, and acquisition metadata are not passed directly to the reranker. Mass and formula constraints affect candidate-pool construction only; no explicit candidate–candidate structural-similarity feature is provided.

The candidate-set Transformer uses a Transformer encoder without candidate positional encodings to exchange information across candidates [22]:

$$[\tilde{\mathbf{h}}_1, \ldots, \tilde{\mathbf{h}}_K] = \mathrm{Transformer}([\mathbf{h}_1, \ldots, \mathbf{h}_K]). \tag{9}$$

Padding masks support variable-sized lists. At deterministic inference, jointly permuting the candidates and their attached embeddings, scores, and ranks permutes the output scores in the same way; the base rank is a candidate-attached retrieval prior, not a tensor-slot position encoding [15,17]. The pointwise control sets $\tilde{\mathbf{h}}_i = \mathbf{h}_i$, retaining the same features, score head, residual path, and training losses while removing learned cross-candidate interaction after the retrieval priors have been constructed. It is not parameter- or capacity-matched because it removes the two Transformer layers: the pointwise and candidate-set variants contain exactly 1,478,690 and 7,783,458 trainable reranker parameters, respectively. A score head predicts a correction Δ_i, and the final score is residual:

$$q_i = \alpha b_i + \Delta_i, \tag{10}$$

where α is initialized to 1.0 and learned without a sign constraint. The residual path provides a direct shortcut from the base score; it does not guarantee preservation of the base order. Moreover, because b_i also enters $\mathbf{x}_i$ and Δ_i can learn additional score-dependent effects, α alone is not a mechanistically interpretable measure of reliance on the base retriever.

4.5 Learning Objective

For the labeled queries $\mathcal{B}_L$ in a minibatch, let V_n be the valid candidate set and y_n the positive index. The primary objective averages listwise cross-entropy over labeled queries:

$$\mathcal{L}_{\mathrm{list}} = -\frac{1}{|\mathcal{B}_L|} \sum_{n \in \mathcal{B}_L} \log \frac{\exp(q_{ny_n})}{\sum_{i \in V_n} \exp(q_{ni})}. \tag{11}$$

We also contrast each positive with every other valid candidate retrieved for that query. The implementation averages over all valid positive–negative pairs in the minibatch:

$$\mathcal{L}_{\mathrm{pair}} = \frac{\sum_{n \in \mathcal{B}_L} \sum_{i \in V_n \setminus \{y_n\}} \log(1 + \exp(q_{ni} - q_{ny_n} + \gamma))}{\sum_{n \in \mathcal{B}_L} (|V_n| - 1)}. \tag{12}$$

The complete objective is

$$\mathcal{L} = \mathcal{L}_{\text{list}} + \lambda\mathcal{L}_{\text{pair}}. \tag{13}$$

The listwise term is a top-one likelihood: it separates the positive from the valid list but does not prescribe a total order among negatives. Both terms are invariant to adding the same query-specific constant to every q_i. Thus, the raw outputs are relative scores, not probabilities; probability calibration is neither performed nor evaluated.

For L Transformer layers and hidden width h, the candidate-set Transformer adds $O(L(Kh^2 + K^2h))$ computation and $O(LK^2)$ attention memory over the pointwise scorer. Neither variant decodes molecules token by token. An empirical latency comparison with GLMR is not reported, so we make no measured efficiency claim.

5 Experiments

5.1 Dataset and Candidate Pools

We use the published MassSpecGym split [3]: 194,119 training spectra, 19,429 validation spectra, and 17,556 test spectra. These counts total 231,104 spectra; the benchmark reports approximately 29,000 unique molecular structures. Its similarity-aware, structure-disjoint construction reduces close structural overlap between training and test molecules.

An exact audit of spectrum-record signatures (peak arrays, precursor m/z, adduct, instrument, and collision energy) found no target-molecule duplicates across splits, but six validation and three test signatures exactly matched training signatures. In all nine matched pairs, the target molecules differ and each evaluation query is marked `simulation_challenge=true`. We report these cases as cross-split spectrum-record overlap. In the overlap-clean alignment-seed-42 pipeline used as our canonical configuration, all validation-dependent model selection excludes the six affected validation spectra, leaving 19,423 validation queries for the rerankers; alignment validation keys decrease from 3,386 to 3,384 molecular entries. We retain and evaluate all 17,556 published test queries.

MassSpecGym supplies mass- and formula-conditioned candidate settings. The mass pool uses a precursor-mass tolerance; the formula pool assumes a known formula and is not unrestricted annotation. Frozen pools contain at most 256 candidates. Before truncation, their test-query means are 253.88 and 165.75, respectively; 1 mass and 3,159 formula pools contain fewer than 40 candidates. After top-40 truncation, the means are 40.00 and 35.73.

Our cache and evaluator assign one positive by exact target-SMILES equality. In official code commit `4f501b3`,[1] the reference loader instead defaults to two-dimensional InChIKey equivalence and may label multiple positives. Our Recall and MRR therefore use a local exact-target-SMILES single-positive protocol on the supplied candidate files; the effect of the identity-rule difference is unquantified.

[1] MassSpecGym reference loader and label transform, accessed August 1, 2026.

5.2 Implementation Details

The aligned model outputs 512-dimensional embeddings. Its spectrum Transformer has four layers, 16 heads, and width 512; the molecular GINE has four layers, hidden width 128, and dropout 0.2. No published SpecEmbedding checkpoint is loaded, so the stage that would freeze a pretrained spectrum tower is skipped; both towers are optimized end to end on the MassSpecGym training split. Alignment uses AdamW with batch size 128, learning rate and weight decay 10^{-4}, cosine scheduling, gradient-norm clipping at 1, at most 100 epochs, and patience 5. Each epoch makes ten draws per molecular label, randomly selecting an associated spectrum. Spectrum and graph augmentation are independently enabled with probability 0.5. The former removes a floor-rounded 20% of fragment peaks below normalized intensity 0.3 and jitters intensities by $\pm15\%$; the latter applies 10% node and 10% directed edge-entry dropout. Validation disables augmentation but retains replicate-spectrum sampling under the fixed random stream. We designate the overlap-clean alignment-seed-42 run and its caches as the canonical configuration. This run applies the validation exclusion above, selects epoch 16, and stops at epoch 21.

The reranker has hidden width 512, a 32-dimensional rank embedding, and dropout 0.1. The candidate-set Transformer has two Transformer layers with eight attention heads; the pointwise control removes both layers. We use AdamW with learning rate 10^{-4}, batch size 16, at most 30 epochs, and early stopping on validation MRR with patience 5. The pairwise weight and margin are $\lambda = 0.2$ and $\gamma = 0.1$. For the canonical controlled analysis, the alignment-seed-42 checkpoint and its overlap-clean caches are fixed while each reranker variant is trained with seeds 42, 43, and 44. These sample standard deviations measure reranker variation conditional on that first stage. A separate cross-alignment sensitivity analysis applies the same reranker seeds to alignment seeds 42, 43, and 44 and first aggregates reranker seeds within each alignment. Its cross-alignment unit is therefore the alignment checkpoint ($n = 3$), not nine flattened runs. The component audit remains limited to the canonical alignment-seed-42 caches and reranker seeds. Its five removals respectively disable only the residual shortcut while retaining the base score as MLP input; jointly disable the base-score feature and shortcut; zero the rank embedding while preserving input width; remove product and absolute-difference features while retaining self-attention; or set $\lambda = 0$ for listwise-only training. All models are selected without test-set access and evaluated on one NVIDIA RTX 4090.

Before training-only positive forcing, top-40 base recall is 93.78% for mass and 90.42% for formula on the training split. Training lists are then completed to 100% labeled coverage as described above. No target is forced into the test top-40 lists, whose corresponding upper bounds are 83.88% and 89.26%.

5.3 Baselines and Metrics

The controlled baselines are the same aligned retriever without the second-stage ranker and a pointwise reranker that removes candidate self-attention while retaining all other inputs and objectives, but not the same parameter count. We

Table 2. Representative local MassSpecGym results. The base uses alignment seed 42, and the learned row uses reranker seed 42. Recall and MRR are percentages under our exact-target-SMILES single-positive protocol on supplied candidate files; they are not official-evaluator-equivalent. MCES@1 is the local thresholded myopic-MCES output (lower is better).

Pool	Method	R@1	R@5	R@20	MRR	MCES@1
Mass	SpecMolAlign (alignment seed 42)	47.46	63.51	77.76	55.03	15.37
	Candidate-set Transformer (rerank seed 42)	68.74	77.77	82.18	72.87	7.71
Formula	SpecMolAlign (alignment seed 42)	63.10	75.79	84.93	68.86	5.44
	Candidate-set Transformer (rerank seed 42)	74.75	82.25	86.89	78.22	3.09

separately show JESTR and GLMR values from Table 1 of the GLMR paper [25]. These external numbers are not a controlled comparison: JESTR and GLMR were not independently reproduced in our codebase, and their retrievers, candidate handling, identity/canonicalization rules, and MCES implementations are not matched to our local exact-target-SMILES single-positive protocol.

We report Recall@1, Recall@5, Recall@20, and mean reciprocal rank (MRR) over all test queries. For every query, including those whose target is absent from the retrieved top-40, let $\hat{m}_n^{(1)}$ be the top-ranked candidate. We report

$$\text{MCES@1} = \frac{1}{N}\sum_{n=1}^{N} \tilde{d}_{15}\left(\hat{m}_n^{(1)}, m_n^{+}\right), \qquad N = 17{,}556. \tag{14}$$

Here $\tilde{d}_{15}$ is the output of `myopic-MCES` with threshold 15 and the stronger-bound option enabled [14]. It is exact for sufficiently close pairs; for distant pairs it may return the threshold or a computed lower bound rather than the unconstrained exact MCES distance. Thus, the reported quantity is thresholded myopic MCES, and lower is better. Exact SMILES-string matches are assigned zero. The canonical runs selected `PULP_CBC_CMD`, and no MCES worker failures were logged. Recall and MRR uncertainty is computed over three reranker seeds with a fixed base checkpoint in the canonical audit. The cross-alignment summary first averages reranker seeds within each checkpoint and then uses the three alignment-level paired estimates; neither sample standard deviation is a confidence interval. MCES@1 is computed for the overlap-clean base and representative seed-42 Transformer checkpoints; these single-checkpoint values do not include seed-level uncertainty.

5.4 Main Results

In the representative seed-42 run, the candidate-set Transformer adds 21.28 percentage points in mass-conditioned Recall@1 and 11.65 points in formula-conditioned Recall@1. The corresponding MRR gains are 17.84 and 9.36 points. Improvements at Recall@20 are smaller because the first-stage top-40 upper

Table 3. External cross-paper context reported by Zhang et al. [25]. These values were not independently reproduced and are not directly comparable to Table 2; retrievers, candidate and identity processing, and MCES implementations may differ.

Pool	Method	R@1	R@5	R@20	MRR	MCES@1
Mass	JESTR (reported; not reproduced)	17.62	40.36	64.76	29.12	15.82
	GLMR (reported; not reproduced)	64.17	72.96	78.78	67.82	11.14
Formula	JESTR (reported; not reproduced)	11.77	33.26	61.01	22.83	11.73
	GLMR (reported; not reproduced)	68.48	78.09	84.22	72.47	5.05

Table 4. Alignment-stratified internal results using MassSpecGym supplied-candidate top-40 lists and our local exact-target-SMILES single-positive evaluation, which is not official-evaluator-equivalent. Learned entries are means over reranker seeds 42–44 within each alignment. The three checkpoints come from two code revisions and provide descriptive sensitivity evidence, not confidence intervals, significance tests, or general end-to-end random-seed stability. Recall@1 and MRR are percentages.

Alignment	Pool	Base R@1	Point. R@1	Trans. R@1	Base MRR	Point. MRR	Trans. MRR
42	Mass	47.46	67.44	68.58	55.03	71.79	72.60
42	Formula	63.10	73.97	74.57	68.86	77.64	78.02
43	Mass	35.92	66.92	67.19	44.33	70.42	70.58
43	Formula	45.08	69.64	69.30	51.85	72.14	71.96
44	Mass	23.91	57.78	57.42	32.26	61.22	60.91
44	Formula	28.73	59.76	59.82	36.87	63.14	63.34

bound is already the dominant constraint. Relative to the corresponding fixed overlap-clean base, the same seed-42 Transformer reduces MCES@1 from 15.3681 to 7.7057 for mass candidates (49.86%) and from 5.4389 to 3.0913 for formula candidates (43.16%).

Table 3 records the values reported in the GLMR paper without ranking those external pipelines against ours. A matched comparison would require rerunning the baselines with the same retriever, candidate lists, identity rule, and metric implementation. The displayed external ordering therefore supports no relative-performance ranking or method attribution.

5.5 Cross-Alignment and Controlled Component Evidence

Table 4 preserves the alignment hierarchy: reranker seeds are first averaged within each first-stage checkpoint. Both learned rankers improve the alignment-matched base in Recall@1 and MRR in all 12 alignment–candidate–learned-model aggregate cells. In contrast, the Transformer–pointwise difference is positive for only two of three alignments for each candidate setting and metric. Across the three alignment-level paired estimates, the mean ± sample-SD differences are 0.35 ± 0.75 Recall@1 points and 0.22 ± 0.56 MRR points for mass, and 0.11 ± 0.47 and 0.13 ± 0.29 points for formula. These are descriptive summaries, not confidence intervals or significance tests. The three checkpoints share the

Table 5. Controlled comparison on fixed alignment-seed-42 caches. The base row is fixed; learned rows report mean ± sample standard deviation over reranker seeds 42–44. Recall@1 and MRR are percentages. Removal rows are separately trained audits of the predefined full configuration, not post-hoc model-selection candidates. All values use the local exact-target-SMILES single-positive protocol and are not official-evaluator-equivalent.

	Mass		Formula	
Method	R@1	MRR	R@1	MRR
Base	47.46	55.03	63.10	68.86
Pointwise	67.44 ± 0.74	71.79 ± 0.66	73.97 ± 0.59	77.64 ± 0.49
Candidate-set Transformer (full)	68.58 ± 0.20	72.60 ± 0.25	74.57 ± 0.28	78.02 ± 0.26
No residual shortcut	68.16 ± 0.06	72.33 ± 0.06	75.01 ± 0.54	78.38 ± 0.42
No base-score input/shortcut	68.07 ± 0.20	72.28 ± 0.18	74.39 ± 0.55	77.93 ± 0.38
No rank embedding	70.12 ± 0.30	73.75 ± 0.28	76.62 ± 0.69	79.38 ± 0.58
Product/absolute-difference removed	68.10 ± 0.85	72.29 ± 0.66	75.76 ± 0.55	78.90 ± 0.38
Listwise only ($\lambda = 0$)	68.91 ± 0.76	72.86 ± 0.46	74.24 ± 0.25	77.71 ± 0.25

same recorded hyperparameters but were produced from two code revisions; per-alignment provenance is retained in the analysis artifacts. Thus, these internal results support supervised residual reranking but do not establish a checkpoint-robust independent benefit of candidate self-attention.

On the canonical alignment-seed-42 checkpoint in Table 5, the candidate-set Transformer exceeds the pointwise model by 1.14/0.60 Recall@1 points and 0.81/0.38 MRR points for mass/formula candidates, respectively. For each of the six pool-by-reranker-seed pairs, both Recall@1 and MRR favor the Transformer within this checkpoint. Because it also has 6.30M more trainable parameters, even this within-checkpoint comparison does not isolate candidate interaction from increased capacity; the cross-alignment direction changes further preclude a checkpoint-robust self-attention claim.

The separately trained removal rows audit the full configuration on the same fixed alignment. The full model does not achieve six-of-six MRR wins against any removal variant. Removing the rank embedding improves both reported metrics in every pool-by-reranker-seed pair; its mean Recall@1/MRR gains are 1.54/1.15 mass points and 2.05/1.36 formula points. The residual shortcut, product/absolute-difference features, and pairwise loss have candidate-dependent directions. The full model's mean MRR lead over the joint base-score-pathway removal is only 0.32/0.08 points, with mixed seed signs. These are descriptive comparisons, not significance tests; the predefined full configuration remains the reporting reference.

On the pre-overlap-clean fixed checkpoint, a sensitivity check excluded the three affected test queries, leaving 17,553. Across both candidate settings and rerankers, mean Recall@1 decreased by 0.0045–0.0055 points (maximum seed-level change: 0.0056); all changes were below 0.006 percentage points. Canonical tables retain all test queries.

The overlap-clean and pre-clean pipelines select the same alignment epochs (best 16, stop 21), but reranker best/stop epochs differ in 9 of 12 configurations. Because alignment was retrained and its test metrics changed, this measures pipeline sensitivity rather than the causal effect of six validation queries.

We do not report candidate-order or full top-K ablations. We also do not report peak memory, cache construction time, reranker latency, or end-to-end latency.

6 Discussion

Equations 2–3 separate first-stage coverage from within-list ordering. For $k = 1$ and $K = 40$, the defined conditional quantity $A_{1|40}$ equals $\mathrm{R@1}/U_{40}$. Using the displayed three-seed means, it increases from 56.58% for the base retriever to 80.40% for the pointwise model and 81.76% for the candidate-set Transformer under mass conditioning. The corresponding formula-conditioned rates are 70.69%, 82.87%, and 83.54%.

Another view is the fraction of attainable top-1 headroom recovered,

$$H = \frac{\mathrm{R@1}_{\mathrm{rerank}} - \mathrm{R@1}_{\mathrm{base}}}{U_{40} - \mathrm{R@1}_{\mathrm{base}}}. \tag{15}$$

The pointwise and candidate-set models recover 54.86% and 57.99% of this headroom for mass candidates, and 41.55% and 43.85% for formula candidates. The pointwise gain is also 94.60% and 94.77% of the total candidate-set gain over the base model in the two settings. Thus, most of the observed improvement is already achieved by the supervised pointwise second-stage ranker as configured; the candidate-set variant adds only a modest increment. At Recall@20, the candidate-set three-seed means of 82.20% and 86.77% are only 1.68 and 2.49 points below U_{40}; first-stage coverage is therefore the dominant remaining limitation for that metric under this top-40 protocol. The removals likewise support only a framework-level interpretation: the full model does not achieve six-of-six MRR wins against any removal variant, whereas rank-embedding removal improves both metrics in every pool-by-reranker-seed pair. The predefined full configuration remains the reporting reference; these algebraic and removal comparisons are not causal estimates of individual components. Across alignment seeds 42–44, both learned variants retain a large descriptive margin over their corresponding base, despite substantial first-stage metric variation. This strengthens the framework-level observation within the three audited runs, while the mixed Transformer–pointwise directions reinforce that most of the gain should not be attributed to candidate self-attention.

7 Limitations and Responsible Evaluation

Three-seed sample standard deviations and paired-win counts are descriptive, not confidence intervals or significance tests. The canonical table and all component removals fix alignment seed 42. The separate cross-alignment analysis covers only three internal checkpoints, produced by two code revisions, and first aggregates reranker seeds within each alignment; it is descriptive sensitivity evidence rather than a confidence interval or general stability claim. The component audit therefore establishes neither cross-alignment component stability nor causal effects. The joint base-score-pathway removal changes both its MLP input and shortcut; product/absolute-difference removal retains self-attention; and the rank-free variant was neither tested across alignments nor selected post hoc. The candidate-set model also has 5.26 times the pointwise model's trainable parameters (6,304,768 more; +426%), confounding interaction with capacity. External JESTR and GLMR rows were not reproduced under identical software and candidate handling; we claim neither definitive state of the art nor an independent component benefit.

Training-only positive replacement changes the list distribution without addressing $1 - U_K$. Because frozen full pools contain the exact target, pool-level evaluation is closed-set even when top-40 retrieval misses it. The impact of exact-target-SMILES rather than reference two-dimensional InChIKey identity is unquantified, so Recall/MRR are explicitly local-protocol results rather than official-evaluator-equivalent values. The exact-record audit cannot exclude near duplicates or correlated provenance. Model selection excludes the six audited training–validation overlaps; all test queries remain. The post-hoc validation environment is now version-locked and was rebuilt from explicit Conda/PyPI locks, but historical training runs did not preserve a per-package environment record; therefore we do not claim bitwise retraining equivalence. A 2026 audit preprint reports further MassSpecGym leakage, shortcut, implementation, and metric risks and releases v1.5 [16]; our scores are artifact- and protocol-specific, not proven v1.5-equivalent.

Alignment sampling approximately balances molecular labels, whereas metrics count spectra, giving replicate-rich molecules more weight. Candidate-pool Recall measures prioritization, not confirmed identification; confidence frameworks and work on spectral-library absences distinguish these notions [19,10]. Conformal retrieval can produce spectrum-specific sets with marginal coverage under exchangeability and target-in-pool assumptions [18], but such guarantees do not automatically cover a target missing from the retrieved top-40. We perform neither probability calibration nor abstention; orthogonal experimental evidence remains necessary.

Finally, formula conditioning presumes a known formula. The spectrum tower uses selected peaks and precursor m/z, but no explicit adduct, instrument, or collision-energy inputs. Robustness to acquisition shift, other retrievers and embeddings, identity rules, pool construction, larger lists, other benchmarks, and external libraries is untested. The closed-set method cannot propose structures outside the supplied pool despite avoiding autoregressive generation-validity failures.

8 Conclusion

We presented SpecEmbedding-Rerank for supervised, non-generative second-stage reranking of MS/MS candidates. Recall and MRR use MassSpecGym-supplied candidate files with a local exact-target-SMILES single-positive protocol. Because the reference loader defaults to two-dimensional InChIKey equivalence and may yield multiple positives, the unquantified difference means these results are not official-evaluator-equivalent. Within this protocol, both learned variants improve their alignment-matched base in both metrics in all 12 aggregate cells across alignment seeds 42–44. The candidate-set Transformer has a modest canonical-checkpoint advantage over the pointwise scorer, but its direction is positive in only two of three alignments for each candidate setting and metric. The full model does not achieve six-of-six MRR wins against any of five removal variants within the canonical alignment, while rank-embedding removal improves both reported metrics in every pool-by-reranker-seed pair. The strongest supported conclusion therefore concerns the reranking framework, not an independent component benefit. The separately displayed JESTR/GLMR values were not reproduced and are not directly comparable; matched external baselines and measured efficiency remain outside the evidence.

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